use, there is no reason why these should be any different from the machinations—all too predictable to a selfish gene/extended phenotype theorist—of viruses.

When we have a cold or a cough, we normally think of the symptoms as annoying byproducts of the virus's activities. But in some cases it seems more probable that they are deliberately engineered by the virus to help it to travel from one host to another. Not content with simply being breathed into the atmosphere, the virus makes us sneeze or cough explosively. The rabies virus is transmitted in saliva when one animal bites another. In dogs, one of the symptoms of the disease is that normally peaceful and friendly animals become ferocious biters, foaming at the mouth. Ominously too, instead of staying within a mile or so of home like normal dogs, they turn into restless wanderers, propagating the virus far afield. It has even been suggested that the well-known hydrophobic symptom encourages the dog to shake the wet foam from its mouth—and with it the virus. I do not know of any direct evidence that sexually transmitted diseases increase the libido of sufferers, but I conjecture that it would be worth looking into. Certainly at least one alleged aphrodisiac, Spanish Fly, is said to work by inducing an itch … and making people itch is just the kind of thing viruses are good at.

The point of comparing rebel human DNA with invading parasitic viruses is that there really isn't any important difference between them. Viruses may well, indeed, have originated as collections of breakaway genes. If we want to erect any distinction, it should be between genes that pass from body to body via the orthodox route of sperms or eggs, and genes that pass from body to body via unorthodox, 'sideways' routes. Both classes may include genes that originated as 'own' chromosomal genes. And both classes may include genes that originated as external, invading parasites. Or perhaps, as I speculated on page 237, all 'own' chromosomal genes should be regarded as mutually parasitic on one another. The important difference between my two classes of genes lies in the divergent circumstances from which they stand to benefit in the future. A cold virus gene and a breakaway human chromosomal gene agree with one another in 'wanting' their host to sneeze. An orthodox chromosomal gene and a venereally transmitted virus agree with one another in wanting their host to copulate. It is an intriguing thought that both would want the host to be sexually attractive. More, an orthodox chromosomal gene and a virus that is transmitted